Practical 8: Implementation of IPSec Site-to-Site VPNs

Aim: Implement IP Sec Site-to-Site VPNs

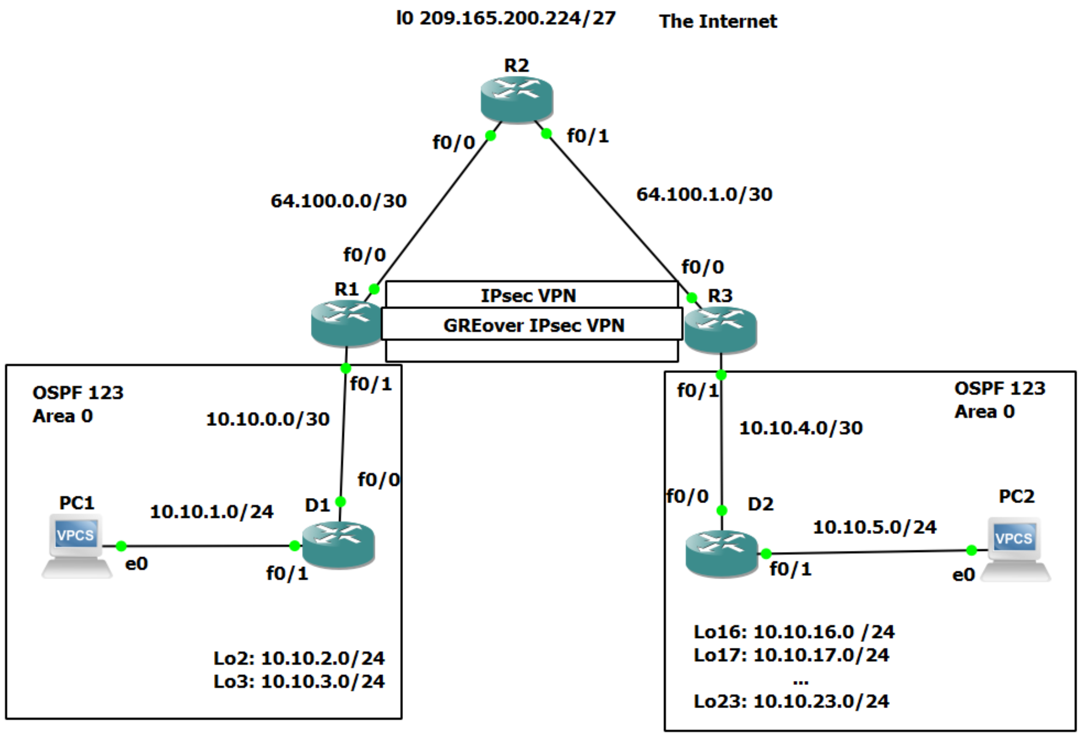
What is IP Sec VPN?

IPsec is a set of protocols that work together to establish encrypted connections between devices. It contributes to the security of data sent over public networks. IPsec is a popular VPN protocol that works by encrypting IP packets and authenticating the source of the packets.

Users can connect to an IPsec VPN by launching a VPN client application This usually necessitates the user having the application installed on their device.

VPN logins are usually password-based While data sent over a VPN is encrypted, if user passwords are compromised, attackers can log into the VPN and steal this encrypted data. Using two-factor authentication (2FA) can strengthen IPsec VPN security, since stealing a password alone will no longer give an attacker access.

Step 1: Design the network topology.



Step 2: Configure the network.

Router 1 (R1):

conf t

no ip domain-lookup

line console 0

```
 logging synchronous
 exec-timeout 0 0
 exit
interface FastEthernet0/0
 description Connection to R2
 ip address 64.100.0.2 255.255.255.252
 shutdown
 no shutdown
 exit
interface FastEthernet0/1
 description Connection to D1
 ip address 10.10.0.1 255.255.255.252
 shutdown
 no shutdown
 exit

router ospf 123
 router-id 1.1.1.1
 auto-cost reference-bandwidth 1000
 network 10.10.0.0 0.0.0.3 area 0
 default-information originate
 exit
ip route 0.0.0.0 0.0.0.0 64.100.0.1
end
wr mem
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip domain-lookup
R1(config)#line console 0
R1(config-line)# logging synchronous
R1(config-line)# exec-timeout 0 0
R1(config-line)# exit
R1(config)#interface FastEthernet0/0
R1(config-if)# description Connection to R2
R1(config-if)# ip address 64.100.0.2 255.255.255.252
R1(config-if)# shutdown
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#interface FastEthernet0/1
R1(config-if)# description Connection to D1
R1(config-if)# ip address 10.10.0.1 255.255.255.252
R1(config-if)# shutdown
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
R1(config)#router ospf 123
R1(config-router)# router-id 1.1.1.1
R1(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
R1(config-router)# network 10.10.0.0 0.0.0.3 area 0
R1(config-router)# default-information originate
R1(config-router)# exit
R1(config)#ip route 0.0.0.0 0.0.0.0 64.100.0.1
R1(config)#end
R1#wr mem
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...

*Nov 19 16:58:38.447: %SYS-5-CONFIG_I: Configured from console by console[OK]
R1#
*Nov 19 16:58:39.835: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Nov 19 16:58:40.747: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Nov 19 16:58:40.935: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1#
*Nov 19 16:58:41.747: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1#
```

Router 2 (R2):

conf t

no ip domain-lookup

line console 0

logging synchronous

exec-timeout 0 0

exit

! This is R2 - Implement GRE over IPSec Site-to-Site VPN

interface FastEthernet0/0

description Connection to R1

ip address 64.100.0.1 255.255.255.252

shutdown

no shutdown

exit

interface FastEthernet0/1

```
 description Connection to R3
 ip address 64.100.1.1 255.255.255.252
 shutdown
 no shutdown
 exit
interface Loopback0
 description Internet simulated address
 ip address 209.165.200.225 255.255.255.224
 shutdown
 no shutdown
 exit
ip route 0.0.0.0 0.0.0.0 Loopback0
ip route 10.10.0.0 255.255.252.0 64.100.0.2
ip route 10.10.4.0 255.255.252.0 64.100.1.2
ip route 10.10.16.0 255.255.248.0 64.100.1.2
end
wr mem
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no ip domain-lookup
R2(config)#line console 0
R2(config-line)# logging synchronous
R2(config-line)# exec-timeout 0 0
R2(config-line)# exit
R2(config)#! This is R2 - Implement GRE over IPSec Site-to-Site VPN
R2(config)#interface FastEthernet0/0
R2(config-if)# description Connection to R1
R2(config-if)# ip address 64.100.0.1 255.255.255.252
R2(config-if)# shutdown
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#interface FastEthernet0/1
R2(config-if)# description Connection to R3
R2(config-if)# ip address 64.100.1.1 255.255.255.252
R2(config-if)# shutdown
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#interface Loopback0
R2(config-if)# description Internet simulated address
R2(config-if)# ip address 209.165.200.225 255.255.255.224
R2(config-if)# shutdown
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#ip route 0.0.0.0 0.0.0.0 Loopback0
%Default route without gateway, if not a point-to-point interface, may impact performance
R2(config)#ip route 10.10.0.0 255.255.252.0 64.100.0.2
R2(config)#ip route 10.10.4.0 255.255.252.0 64.100.1.2
R2(config)#ip route 10.10.16.0 255.255.248.0 64.100.1.2
R2(config)#end
R2#wr mem
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Nov 19 17:00:12.895: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
*Nov 19 17:00:14.055: %SYS-5-CONFIG_I: Configured from console by console
[confirm]
*Nov 19 17:00:14.247: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Nov 19 17:00:14.731: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Nov 19 17:00:15.247: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
[confirm]
*Nov 19 17:00:15.731: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
[confirm]
Building configuration...
[OK]
R2#
```

Router 3 (R3):

conf t

no ip domain-lookup

line console 0

logging synchronous

exec-timeout 0 0

exit

! This is R3 - Implement GRE over IPSec Site-to-Site VPN

interface FastEthernet0/0

description Connection to R2

ip address 64.100.1.2 255.255.255.252

shutdown

no shutdown

```
 exit
interface FastEthernet0/1
 description Connection to D2
 ip address 10.10.4.1 255.255.255.252
 shutdown
 no shutdown
 exit
ip route 0.0.0.0 0.0.0.0 64.100.1.1
router ospf 123
 router-id 3.3.3.1
 auto-cost reference-bandwidth 1000
 network 10.10.4.0 0.0.0.3 area 0
 default-information originate
 exit
end
wr mem
```

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#no ip domain-lookup
R3(config)#line console 0
R3(config-line)# logging synchronous
R3(config-line)# exec-timeout 0 0
R3(config-line)# exit
R3(config)#! This is R3 - Implement GRE over IPSec Site-to-Site VPN
R3(config)#interface FastEthernet0/0
R3(config-if)# description Connection to R2
R3(config-if)# ip address 64.100.1.2 255.255.255.252
R3(config-if)# shutdown
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#interface FastEthernet0/1
R3(config-if)# description Connection to D2
R3(config-if)# ip address 10.10.4.1 255.255.255.252
R3(config-if)# shutdown
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#ip route 0.0.0.0 0.0.0.0 64.100.1.1
R3(config)#router ospf 123
R3(config-router)# router-id 3.3.3.1
R3(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
R3(config-router)# network 10.10.4.0 0.0.0.3 area 0
R3(config-router)# default-information originate
R3(config-router)# exit
R3(config)#end
R3#wr mem
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Nov 19 17:01:20.763: %SYS-5-CONFIG_I: Configured from console by console
[confirm]
*Nov 19 17:01:21.531: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Nov 19 17:01:21.987: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Nov 19 17:01:22.531: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
[confirm]
*Nov 19 17:01:22.987: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
[confirm]
Building configuration...
[OK]
R3#
```

• **Router 4 (D1):**

conf t

no ip domain-lookup

line console 0

logging synchronous

exec-timeout 0 0

exit

! This is D1 - Implement GRE over IPSec Site-to-Site VPN

interface FastEthernet0/0

description Connection to R1

ip address 10.10.0.2 255.255.255.252

shutdown

no shutdown

exit

```
interface FastEthernet0/1
 description Connection to PC1
 ip address 10.10.1.1 255.255.255.0
 shutdown
 no shutdown
 exit
interface Loopback2
 description Loopback to simulate an OSPF network
 ip address 10.10.2.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit

interface Loopback3
 description Loopback to simulate an OSPF network
 ip address 10.10.3.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
router ospf 123
 router-id 1.1.1.2
 auto-cost reference-bandwidth 1000
 network 10.10.0.0 0.0.3.255 area 0
 exit
end
wr mem
```

```
D1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#no ip domain-lookup
D1(config)#line console 0
D1(config-line)# logging synchronous
D1(config-line)# exec-timeout 0 0
D1(config-line)# exit
D1(config)#! This is D1 - Implement GRE over IPSec Site-to-Site VPN
D1(config)#interface FastEthernet0/0
D1(config-if)# description Connection to R1
D1(config-if)# ip address 10.10.0.2 255.255.255.252
D1(config-if)# shutdown
D1(config-if)# no shutdown
D1(config-if)# exit
D1(config)#interface FastEthernet0/1
D1(config-if)# description Connection to PC1
D1(config-if)# ip address 10.10.1.1 255.255.255.0
D1(config-if)# shutdown
D1(config-if)# no shutdown
D1(config-if)# exit
D1(config)#interface Loopback2
D1(config-if)# description Loopback to simulate an OSPF network
D1(config-if)# ip address 10.10.2.1 255.255.255.0
D1(config-if)# ip ospf network point-to-point
D1(config-if)# shutdown
D1(config-if)# no shutdown
D1(config-if)# exit
D1(config)#
D1(config)#interface Loopback3
D1(config-if)# description Loopback to simulate an OSPF network
D1(config-if)# ip address 10.10.3.1 255.255.255.0
D1(config-if)# ip ospf network point-to-point
D1(config-if)# shutdown
D1(config-if)# no shutdown
D1(config-if)# exit
D1(config)#router ospf 123
D1(config-router)#
*Nov 19 17:02:54.875: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up
*Nov 19 17:02:55.655: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback3, changed state to up router-id 1.1.1.2
D1(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
D1(config-router)# network 10.10.0.0 0.0.3.255 area 0
D1(config-router)# exit
D1(config)#end
D1#wr mem
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Nov 19 17:02:56.339: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Nov 19 17:02:56.695: %SYS-5-CONFIG_I: Configured from console by console
```

- **Router 5 (D2):**

conf t

no ip domain-lookup

line console 0

logging synchronous

exec-timeout 0 0

exit

! This is D2 - Implement GRE over IPSec Site-to-Site VPN

interface FastEthernet0/0

description Connection to R3

ip address 10.10.4.2 255.255.255.252

shutdown

no shutdown

exit

```
interface FastEthernet0/1
 description Connection to PC2
 ip address 10.10.5.1 255.255.255.0
 shutdown
 no shutdown
 exit
interface Loopback16
 description Loopback to simulate an OSPF network
 ip address 10.10.16.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
interface Loopback17
 description Loopback to simulate an OSPF network
 ip address 10.10.17.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
interface Loopback18
 description Loopback to simulate an OSPF network
 ip address 10.10.18.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
interface Loopback19
 description Loopback to simulate an OSPF network
 ip address 10.10.19.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
```

```
interface Loopback20
 description Loopback to simulate an OSPF network
 ip address 10.10.20.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
interface Loopback21
 description Loopback to simulate an OSPF network
 ip address 10.10.21.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
interface Loopback22
 description Loopback to simulate an OSPF network
 ip address 10.10.22.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
interface Loopback23
 description Loopback to simulate an OSPF network
 ip address 10.10.23.1 255.255.255.0
 ip ospf network point-to-point
 shutdown
 no shutdown
 exit
router ospf 123
 router-id 3.3.3.2
 auto-cost reference-bandwidth 1000
 network 10.10.4.0 0.0.1.255 area 0
 network 10.10.16.0 0.0.7.255 area 0
 exit
```

end

wr mem

```
D2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)#no ip domain-lookup
D2(config)#line console 0
D2(config-line)# logging synchronous
D2(config-line)# exec-timeout 0 0
D2(config-line)# exit
D2(config)#! This is D2 - Implement GRE over IPSec Site-to-Site VPN
D2(config)#interface FastEthernet0/0
D2(config-if)# description Connection to R3
D2(config-if)# ip address 10.10.4.2 255.255.255.252
D2(config-if)# shutdown
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#interface FastEthernet0/1
D2(config-if)# description Connection to PC2
D2(config-if)# ip address 10.10.5.1 255.255.255.0
D2(config-if)# shutdown
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#interface Loopback16
D2(config-if)# description Loopback to simulate an OSPF network
D2(config-if)# ip address 10.10.16.1 255.255.255.0
D2(config-if)# ip ospf network point-to-point
D2(config-if)# shutdown
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#interface Loopback17
D2(config-if)# description Loopback to simulate an OSPF network
D2(config-if)# ip address 10.10.17.1 255.255.255.0
D2(config-if)#
*Nov 19 17:04:32.367: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback16, changed state to up
*Nov 19 17:04:33.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback17, changed state to up ip ospf network point-to-point
D2(config-if)# shutdown
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#interface Loopback18
D2(config-if)# description Loopback to simulate an OSPF network
D2(config-if)# ip address 10.10.18.1 255.255.255.0
D2(config-if)# ip ospf network point-to-point
D2(config-if)# shutdown
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#interface Loopback19
D2(config-if)# description Loopback to simulate an OSPF ne
*Nov 19 17:04:33.787: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback18, changed state to up
*Nov 19 17:04:33.811: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Nov 19 17:04:34.203: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to uptwork
D2(config-if)# ip address 10.10.19.1 255.255.255.0
D2(config-if)# ip ospf network point-to-point
D2(config-if)# shutdown
```

```
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#interface Loopback20
D2(config-if)# description Loopback to simulate an OSPF network
D2(config-if)# ip address 10.10.20.1 255.255.255.0
D2(config-if)# ip ospf network point-to-point
D2(config-if)# shutdown
D2(config-if)#
*Nov 19 17:04:34.403: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback19, changed state to up
*Nov 19 17:04:34.819: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
*Nov 19 17:04:35.051: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback20, changed state to up
*Nov 19 17:04:35.227: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up no shutdown
D2(config-if)# exit
D2(config)#interface Loopback21
D2(config-if)# description Loopback to simulate an OSPF network
D2(config-if)# ip address 10.10.21.1 255.255.255.0
D2(config-if)# ip ospf network point-to-point
D2(config-if)# shutdown
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#interface Loopback22
D2(config-if)# description Loopback to simulate an OSPF network
D2(config-if)# ip address 10.10.22.1 255.255.255.0
D2(config-if)# ip ospf network point-to-point
D2(config-if)# shutdown
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#interface Loopback23
D2(config-if)# description Loopback to simulate an OSPF network
D2(config-if)# ip address 10.10.23.1 255.255.255.0
D2(config-if)# ip ospf network point-to-point
D2(config-if)# shutdown
D2(config-if)# no shutdown
D2(config-if)# exit
D2(config)#router ospf 123
D2(config-router)#
*Nov 19 17:04:35.723: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback21, changed state to up
*Nov 19 17:04:36.339: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback22, changed state to up router-id 3.3.3.2
D2(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
D2(config-router)# network 10.10.4.0 0.0.1.255 area 0
D2(config-router)# network 10.10.16.0 0.0.7.255 area 0
D2(config-router)# exit
D2(config)#end
D2#wr mem
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Nov 19 17:04:36.951: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback23, changed state to up
[confirm]
*Nov 19 17:04:38.191: %SYS-5-CONFIG_I: Configured from console by console
```

PC 1:

PC1> ip 10.10.1.10/24 10.10.1.1

Checking for duplicate address...

PC1: 10.10.1.10 255.255.255.0 gateway 10.10.1.1

PC1> sh ip

```
PC1> ip 10.10.1.10/24 10.10.1.1
Checking for duplicate address...
PC1 : 10.10.1.10 255.255.255.0 gateway 10.10.1.1

PC1> sh ip

NAME        : PC1[1]
IP/MASK     : 10.10.1.10/24
GATEWAY     : 10.10.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10032
RHOST:PORT  : 127.0.0.1:10033
MTU:        : 1500

PC1>
```

PC2:

PC2> ip 10.10.5.10/24 10.10.5.1

PC1: 10.10.5.10 255.255.255.0 gateway 10.10.5.1

PC2> sh ip

```
PC2> ip 10.10.5.10/24 10.10.5.1
Checking for duplicate address...
PC1 : 10.10.5.10 255.255.255.0 gateway 10.10.5.1

PC2> sh ip

NAME        : PC2[1]
IP/MASK     : 10.10.5.10/24
GATEWAY     : 10.10.5.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 10034
RHOST:PORT  : 127.0.0.1:10035
MTU:        : 1500

PC2>
```

Step 3: On PC1, verify end-to-end connectivity.

From PC1, ping the first loopback on D3 (10.10.16.1).

PC1> ping 10.10.16.1

```
PC1> ping 10.10.16.1
84 bytes from 10.10.16.1 icmp_seq=1 ttl=251 time=213.351 ms
84 bytes from 10.10.16.1 icmp_seq=2 ttl=251 time=216.580 ms
84 bytes from 10.10.16.1 icmp_seq=3 ttl=251 time=231.729 ms
84 bytes from 10.10.16.1 icmp_seq=4 ttl=251 time=197.569 ms
84 bytes from 10.10.16.1 icmp_seq=5 ttl=251 time=184.142 ms
```

Finally, from PC1, ping the default gateway loopback on R2 (209.165.200.225).

PC1> ping 209.165.200.225

```
PC1> ping 209.165.200.225
84 bytes from 209.165.200.225 icmp_seq=1 ttl=253 time=157.364 ms
84 bytes from 209.165.200.225 icmp_seq=2 ttl=253 time=100.375 ms
84 bytes from 209.165.200.225 icmp_seq=3 ttl=253 time=116.564 ms
84 bytes from 209.165.200.225 icmp_seq=4 ttl=253 time=100.379 ms
84 bytes from 209.165.200.225 icmp_seq=5 ttl=253 time=98.123 ms
```

Step 4: Verify the routing table of R1 and R3.

Verify the OSPF routing table of R1.

R1#sh ip route ospf

```
R1#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 64.100.0.1 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
O        10.10.1.0/24 [110/20] via 10.10.0.2, 00:06:44, FastEthernet0/1
O        10.10.2.0/24 [110/11] via 10.10.0.2, 00:06:44, FastEthernet0/1
O        10.10.3.0/24 [110/11] via 10.10.0.2, 00:06:44, FastEthernet0/1
```

Verify the routing table of R3.

R3#sh ip route ospf

```
R3#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 64.100.1.1 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 11 subnets, 3 masks
O        10.10.5.0/24 [110/20] via 10.10.4.2, 00:05:37, FastEthernet0/1
O        10.10.16.0/24 [110/11] via 10.10.4.2, 00:05:37, FastEthernet0/1
O        10.10.17.0/24 [110/11] via 10.10.4.2, 00:05:37, FastEthernet0/1
O        10.10.18.0/24 [110/11] via 10.10.4.2, 00:05:37, FastEthernet0/1
O        10.10.19.0/24 [110/11] via 10.10.4.2, 00:05:37, FastEthernet0/1
O        10.10.20.0/24 [110/11] via 10.10.4.2, 00:05:37, FastEthernet0/1
O        10.10.21.0/24 [110/11] via 10.10.4.2, 00:05:37, FastEthernet0/1
O        10.10.22.0/24 [110/11] via 10.10.4.2, 00:05:37, FastEthernet0/1
O        10.10.23.0/24 [110/11] via 10.10.4.2, 00:05:37, FastEthernet0/1
R3#
```

Step 5: Configure GRE over IPsec using a Crypto Map on R1.

• On R1, configure the ISAKMP policy and pre-shared key.

Like site-to-site VPNs using crypto maps, GRE over IPsec also requires an ISAKMP policy configuration and pre-shared key configured.

In this lab, we will use the following parameters for the ISAKMP policy 10 on R1:

o Encryption: aes 256

o Hash: sha256

o Authentication method: pre-share key

o Diffie-Hellman group: 14

o Lifetime: 3600 seconds (60 minutes / 1 hour)

Configure ISAKMP policy 10 on R1:

conf t

! Configure ISAKMP (Phase 1) Policy for VPN

crypto isakmp policy 10

encryption aes 256

hash sha

authentication pre-share

group 1

lifetime 36000

exit

end

wr mem

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#! Configure ISAKMP (Phase 1) Policy for VPN
R1(config)#crypto isakmp policy 10
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# hash sha
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 1
R1(config-isakmp)# lifetime 36000
R1(config-isakmp)# exit
R1(config)#end
R1#wr mem
Building configuration...
```

Configure the pre-shared key of cisco123 on R1. This command points to the remote peer R3 G0/0/0 IP address.

R1(config)#

R1(config)#crypto isakmp key cisco123 address 64.100.1.2

R1(config)#

```
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#crypto isakmp key cisco123 address 64.100.1.2
R1(config)#
```

On R1, configure the transform set and VPN ACL.

Create a transform set called GRE-VPN using AES 256 cipher with ESP and the SHA 256 hash function.

R1(config)#crypto ipsec transform-set GRE-VPN esp-aes 256 esp-sha-hmac

R1(cfg-crypto-trans)#

```
R1(config)#crypto ipsec transform-set GRE-VPN esp-aes 256 esp-sha-hmac
R1(cfg-crypto-trans)#
```

Unlike a site-to-site IPsec VPN, the transform must use transport mode. The mode command is used to identify the type of tunnel that will be established. The default is mode tunnel mode. However, GRE over IPsec should be configured using the mode transport command.

R1(cfg-crypto-trans)#mode transport

R1(cfg-crypto-trans)#exit

R1(config)#

```
R1(cfg-crypto-trans)#
R1(cfg-crypto-trans)#mode transport
R1(cfg-crypto-trans)#exit
R1(config)#
```

Next, create a named extended ACL called GRE-VPN-ACL that makes the tunnel interface traffic interesting.

conf t

ip access-list extended GRE-VPN-ACL

permit gre host 64.100.0.2 host 64.100.1.2

exit

end

wr mem

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip access-list extended GRE-VPN-ACL
R1(config-ext-nacl)# permit gre host 64.100.0.2 host 64.100.1.2
R1(config-ext-nacl)# exit
R1(config)#end
R1#wr mem
```

On R1, configure the crypto map and apply it to the interface.

Create a crypto map called GRE-CMAP that associates the new GRE-VPN-ACL, transform set, and peer.

conf t

crypto map GRE-CMAP 10 ipsec-isakmp

match address GRE-VPN-ACL

set transform-set GRE-VPN

set peer 64.100.1.2

exit

end

wr mem

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#crypto map GRE-CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)# match address GRE-VPN-ACL
R1(config-crypto-map)# set transform-set GRE-VPN
R1(config-crypto-map)# set peer 64.100.1.2
R1(config-crypto-map)# exit
R1(config)#end
R1#wr mem
```

Finally, assign a crypto map called GRE-MAP on G0/0/0

conf t

! Apply the crypto map to the outbound interface

interface FastEthernet0/0

crypto map GRE-CMAP

shutdown

no shutdown

exit

end

wr mem

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#! Apply the crypto map to the outbound interface
R1(config)#interface FastEthernet0/0
R1(config-if)# crypto map GRE-CMAP
R1(config-if)# shutdown
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#end
R1#wr mem
```

On R1, configure the GRE tunnel interface.

Configure a GRE tunnel interface as shown. To enable GRE on the tunnel interface, the tunnel gre ipv4 command is required. However, this command is enabled by default and will therefore be configured in our example.

conf t

! Configure GRE Tunnel interface

interface Tunnel1

bandwidth 4000

ip address 172.16.1.1 255.255.255.252

ip mtu 1400

tunnel source 64.100.0.2

tunnel destination 64.100.1.2

shutdown

no shutdown

exit

end

wr mem

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#! Configure GRE Tunnel interface
R1(config)#interface Tunnel1
R1(config-if)# bandwidth 4000
R1(config-if)# ip address 172.16.1.1 255.255.255.252
R1(config-if)# ip mtu 1400
R1(config-if)# tunnel source 64.100.0.2
R1(config-if)# tunnel destination 64.100.1.2
R1(config-if)# shutdown
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#end
R1#wr mem
Building configuration...
```

Step 6: Configure GRE over IPsec using a Tunnel IPsec Profile on R3.

In this part, we will configure GRE over IPsec using tunnel IPsec profiles on R3.

On R3, configure the ISAKMP policy, pre-shared key, and transform set.

In this step, we will configure the same parameters for the ISAKMP policy 10 that we configured on

.

Configure ISAKMP policy 10 on **R3**:

conf t

! Configure ISAKMP (Phase 1) Policy for GRE over IPSec VPN

crypto isakmp policy 10

encryption aes 256

hash sha

authentication pre-share

group 1

lifetime 36000

exit

end

wr mem

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#! Configure ISAKMP (Phase 1) Policy for GRE over IPSec VPN
R3(config)#crypto isakmp policy 10
R3(config-isakmp)# encryption aes 256
R3(config-isakmp)# hash sha
R3(config-isakmp)# authentication pre-share
R3(config-isakmp)# group 1
R3(config-isakmp)# lifetime 36000
R3(config-isakmp)# exit
R3(config)#end
R3#wr mem
```

Configure the pre-shared key of cisco123 on R1. This command points to the remote peer R3 G0/0/0 IP address.

R3(config)#

R3 (config)#crypto isakmp key cisco123 address 64.100.0.2

R3(config)#

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#crypto isakmp key cisco123 address 64.100.0.2
R3(config)#
```

Create a new transform set called GRE-VPN using the same security parameters and transport mode that we configured on R1. Also configure the mode transport command.

conf t

crypto ipsec transform-set GRE-VPN esp-aes 256 esp-sha-hmac

mode transport

exit

end

wr mem

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#crypto ipsec transform-set GRE-VPN esp-aes 256 esp-sha-hmac
R3(cfg-crypto-trans)# mode transport
R3(cfg-crypto-trans)# exit
R3(config)#end
R3#wr mem
Building configuration...
```

On R3, configure the IPsec profile.

Instead of a crypto map, we will configure an IPsec profile called GRE-PROFILE using the crypto ipsec profile ipsec-profile-name global configuration command.

R3 (config)#crypto ipsec profile GRE-profile

R3(ipsec-profile) #

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#crypto ipsec profile GRE-profile
R3(ipsec-profile)#
```

In IPsec profile configuration mode, specify the transform set to be negotiated using the set transform-set transform-set-name command. Multiple transform sets can be specified in order of priority. The fist transform-set-name specified is the highest priority. R3(ipsec-profile)#

R3(ipsec-profile) #set transform-set GRE-VPN

R3(ipsec-profile)#exit

R3(config)#

```
R3(ipsec-profile)#set transform-set GRE-VPN
R3(ipsec-profile)#exit
R3(config)#
```

On R3, configure the tunnel interface.

On R3, configure a GRE tunnel interface.

conf t

! Configure GRE Tunnel interface

interface Tunnel1

bandwidth 4000

ip address 172.16.1.2 255.255.255.252

ip mtu 1400

tunnel source 64.100.1.2

tunnel destination 64.100.0.2

shutdown

no shutdown

exit

end

wr mem

```
R3(config)#interface Tunnel1
R3(config-if)# bandwidth 4000
R3(config-if)# ip address 172.16.1.2 255.255.255.252
R3(config-if)# ip mtu 1400
R3(config-if)# tunnel source 64.100.1.2
R3(config-if)# tunnel destination 64.100.0.2
R3(config-if)# shutdown
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#end
R3#wr mem
Building configuration...

*Nov 19 17:23:47.511: %SYS-5-CONFIG_I: Configured from console by console
*Nov 19 17:23:51.315: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up[OK]
R3#
```

Apply the IPsec profile GRE-PROFILE to the Tunnel 1 interface using the tunnel protection ipsec profile profile-name command.

conf t

interface Tunnel1

tunnel protection ipsec profile GRE-profile

shutdown

no shutdown

exit

end

wr mem

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface Tunnel1
R3(config-if)# tunnel protection ipsec profile GRE-profile
R3(config-if)# shutdown
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#end
R3#wr mem
Building configuration...

*Nov 19 17:24:28.355: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
*Nov 19 17:24:28.511: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Nov 19 17:24:28.667: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
*Nov 19 17:24:28.771: %SYS-5-CONFIG_I: Configured from console by console
*Nov 19 17:24:31.859: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to down[OK]
R3#
*Nov 19 17:24:38.639: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
R3#
```

On R1 and R3, enable OSPF routing on the tunnel interface.

Verify that the GRE over IPsec VPN is operational.

On R1, perform an extended ping to the R3 10.10.16.1 interface

R3#

R1#ping 10.10.16.1 source 10.10.0.1

```
R1#ping 10.10.16.1 source 10.10.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.16.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.0.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 96/158/292 ms
R1#
```

The pings are successful, and it appears that the VPN is operational. On R1, verify the IPsec SA encrypted and decrypted statistics.

show crypto ipsec sa | include encrypt|decrypt

```
R1#show crypto ipsec sa | include encrypt|decrypt
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
R1#
```

From D1, trace the path taken to the R3 10.10.16.1 interface.

D1#trace 10.10.16.1

```
D1#trace 10.10.16.1
Type escape sequence to abort.
Tracing the route to 10.10.16.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.0.1 60 msec 24 msec 32 msec
  2 64.100.0.1 56 msec 60 msec 60 msec
  3 64.100.1.2 88 msec 104 msec 80 msec
  4 10.10.4.2 108 msec 124 msec 116 msec
D1#
```

On R1, configure OSPF to advertise the tunnel interfaces.

conf t

router ospf 123

network 172.16.1.0 0.0.0.3 area 0

exit

end

wr mem

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 123
R1(config-router)# network 172.16.1.0 0.0.0.3 area 0
R1(config-router)# exit
R1(config)#end
R1#wr mem
Building configuration...
```

On R3, configure OSPF to advertise the tunnel interfaces.

conf t

! Enable OSPF on the GRE tunnel network

router ospf 123

network 172.16.1.0 0.0.0.3 area 0

exit

end

wr mem

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#! Enable OSPF on the GRE tunnel network
R3(config)#router ospf 123
R3(config-router)# network 172.16.1.0 0.0.0.3 area 0
R3(config-router)# exit
R3(config)#end
R3#wr mem
Building configuration...
[OK]
R3#
*Nov 19 17:34:04.459: %SYS-5-CONFIG_I: Configured from console by console
*Nov 19 17:34:07.047: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on Tunnel1 from LOADING to FULL, Loading Done
R3#
```

Step 7: Verify the GRE over IPsec Tunnel on R1 and R3

Now that the GRE over IPsec has been configured, we must verify that the tunnel interfaces are correctly enabled, that the crypto session is active, and then generate traffic to confirm it is traversing securely over the IPsec tunnel.

On R1 and R3, verify the tunnel interfaces.

Use the show interfaces tunnel 1 command to verify the interface settings

R1#sh interfaces tunnel 1

```
R1#sh interfaces tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 172.16.1.1/30
  MTU 17916 bytes, BW 4000 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 64.100.0.2, destination 64.100.1.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:08, output 00:00:08, output hang never
  Last clearing of "show interface" counters 00:16:31
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     14 packets input, 1836 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     19 packets output, 2200 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

On R3, use the show interfaces tunnel 1 command to verify the interface settings.

show interface tunnel1 | include is up | Internet address | Enc | Tunnel protocol

```
R3#$ace tunnel1 | include is up | Internet address | Enc | Tunnel protocol
Tunnel1 is up, line protocol is up
  Internet address is 172.16.1.2/30
  Tunnel protocol/transport GRE/IP
R3#
```

On R1 and R3, verify the crypto settings.

On R1, use the show crypto session command to verify the operation of the VPN tunnel.

R1#sh crypto session

```
R1#sh crypto session
Crypto session current status

Interface: FastEthernet0/0
Session status: UP-ACTIVE
Peer: 64.100.1.2 port 500
  IKEv1 SA: local 64.100.0.2/500 remote 64.100.1.2/500 Active
  IPSEC FLOW: permit 47 host 64.100.0.2 host 64.100.1.2
        Active SAs: 2, origin: crypto map
```

On R3, use the show crypto session command to verify the operation of the VPN tunnel. R3#

R3#sh crypto session

```
R3#sh crypto session
Crypto session current status

Interface: Tunnel1
Session status: UP-ACTIVE
Peer: 64.100.0.2 port 500
  IKEv1 SA: local 64.100.1.2/500 remote 64.100.0.2/500 Active
  IPSEC FLOW: permit 47 host 64.100.1.2 host 64.100.0.2
        Active SAs: 2, origin: crypto map
```

On R1 and R3, verify OSPF routing.

On R1 and R3, verify which interfaces are configured for OSPF using the show ip ospf interface brief command.

R1#

R1#sh ip ospf int bri

```
R1#sh ip ospf int bri
Interface    PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Tu1          123   0               172.16.1.1/30      250   P2P   1/1
Fa0/1        123   0               10.10.0.1/30       10    DR    1/1
R1#
```

R3#sh ip ospf int bri

```
R3#sh ip ospf int bri
Interface    PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Tu1          123   0               172.16.1.2/30      250   P2P   1/1
Fa0/1        123   0               10.10.4.1/30       10    DR    1/1
```

On R1 and R3, verify the OSPF neighbours using the show ip ospf interface brief command.

R3#sh ip ospf neighbor

```
R3#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:36    172.16.1.1      Tunnel1
3.3.3.2           1   FULL/BDR        00:00:35    10.10.4.2       FastEthernet0/1
```

R1#sh ip route ospf

```
R1#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 64.100.0.1 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 15 subnets, 3 masks
O        10.10.1.0/24 [110/20] via 10.10.0.2, 00:37:39, FastEthernet0/1
O        10.10.2.0/24 [110/11] via 10.10.0.2, 00:37:39, FastEthernet0/1
O        10.10.3.0/24 [110/11] via 10.10.0.2, 00:37:39, FastEthernet0/1
O        10.10.4.0/30 [110/260] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.5.0/24 [110/270] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.16.0/24 [110/261] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.17.0/24 [110/261] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.18.0/24 [110/261] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.19.0/24 [110/261] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.20.0/24 [110/261] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.21.0/24 [110/261] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.22.0/24 [110/261] via 172.16.1.2, 00:06:40, Tunnel1
O        10.10.23.0/24 [110/261] via 172.16.1.2, 00:06:40, Tunnel1
```

R3#sh ip route ospf

```
R3#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 64.100.1.1 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 15 subnets, 3 masks
O        10.10.0.0/30 [110/260] via 172.16.1.1, 00:07:23, Tunnel1
O        10.10.1.0/24 [110/270] via 172.16.1.1, 00:07:23, Tunnel1
O        10.10.2.0/24 [110/261] via 172.16.1.1, 00:07:23, Tunnel1
O        10.10.3.0/24 [110/261] via 172.16.1.1, 00:07:23, Tunnel1
O        10.10.5.0/24 [110/20] via 10.10.4.2, 00:36:30, FastEthernet0/1
O        10.10.16.0/24 [110/11] via 10.10.4.2, 00:36:30, FastEthernet0/1
O        10.10.17.0/24 [110/11] via 10.10.4.2, 00:36:30, FastEthernet0/1
O        10.10.18.0/24 [110/11] via 10.10.4.2, 00:36:30, FastEthernet0/1
O        10.10.19.0/24 [110/11] via 10.10.4.2, 00:36:30, FastEthernet0/1
O        10.10.20.0/24 [110/11] via 10.10.4.2, 00:36:30, FastEthernet0/1
O        10.10.21.0/24 [110/11] via 10.10.4.2, 00:36:30, FastEthernet0/1
O        10.10.22.0/24 [110/11] via 10.10.4.2, 00:36:30, FastEthernet0/1
O        10.10.23.0/24 [110/11] via 10.10.4.2, 00:36:30, FastEthernet0/1
```

Verify that there is an operational logical point-to-point link between R1 and R3 using the GRE tunnel interface.

R1#sh ip route 172.16.0.0

```
R1#sh ip route 172.16.0.0
Routing entry for 172.16.0.0/16, 2 known subnets
  Attached (2 connections)
  Variably subnetted with 2 masks
C        172.16.1.0/30 is directly connected, Tunnel1
L        172.16.1.1/32 is directly connected, Tunnel1
```

R3#sh ip route 172.16.0.0

```
R3#sh ip route 172.16.0.0
Routing entry for 172.16.0.0/16, 2 known subnets
  Attached (2 connections)
  Variably subnetted with 2 masks
C        172.16.1.0/30 is directly connected, Tunnel1
L        172.16.1.2/32 is directly connected, Tunnel1
```

Test the GRE over IPsec VPN tunnel.

From D1, trace the path taken to the R3 10.10.16.1 interface.

D1#trace 10.10.16.1

```
D1#trace 10.10.16.1
Type escape sequence to abort.
Tracing the route to 10.10.16.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.0.1 12 msec 132 msec 84 msec
  2 172.16.1.2 116 msec 100 msec 80 msec
  3 10.10.4.2 132 msec 116 msec 120 msec
```

On R1, verify the IPsec SA encrypted and decrypted statistics.

R1#sh crypto ipsec sa | include encrypt | decrypt

```
R1#sh crypto ipsec sa | include encrypt | decrypt
    #pkts decaps: 91, #pkts decrypt: 91, #pkts verify: 91
R1#
```

The output verifies that the GRE over IPsec VPN tunnel is properly encrypting traffic between both sites. The packets encrypted include the trace packets along with OSPF packet